Cayleb Alvarez-James

AI Agent Systems Engineer - Autonomous Agents, Agent Harnesses & Self-Improving Infrastructure

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Summary

AI agent systems engineer and technical founder with a year in agentic systems and harness engineering, beginning while finishing high school. I work across several harnesses instead of just one: OpenCode/DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex and Claude Code - routing work by available usage and fit; the current centre is Muse, Claude Duet and the DSH Fleet I am finishing. Three systems define the work: Ergon (self-configuring employee agents), the DSH Fleet (distributed, self-improving dynamic-swarm orchestration) and OpenBell (tiny models built and evaluated in weight space). 1,200+ GitHub contributions in the last year; 4,000+ automated tests across five shipped systems.

Experience

Founder - Independent AI Agent Systems | Oct 2025-present

Architected Ergon, the DSH Fleet and OpenBell: self-configuring agents, dynamic-swarm infrastructure, weight-space model evaluation.

Selected Projects

Ergon (Node.js, C#, Python, SQLite)

- Self-configuring agent that learns each employee's role on its own. In development; 622 tests pass.

DSH Fleet (Python, TypeScript)

- Distributed, self-improving infrastructure: dynamic swarms plus the Omni Loop.

OpenBell (Python, Rust)

- Weight-space training: directly authored tiny specialist models, frozen before scoring.

Sophos (React, TypeScript, Tauri)

- Coding-agent desktop app; sessions survive restarts. 1,224 interface tests; merged after full CI.

dotz (Rust, Tauri, JavaScript, SQLite)

- Runs and supervises AI coding agents. 828+ core tests per platform; all 6 hosted jobs pass; merged.

Asmodeus (Rust)

- Tests trading ideas on past data in time order; no peeking, no real money. 1,111 Rust tests pass.

Industry AI Suite (Python)

- Ten local review tools; every finding shows its source and a person decides. 382 Python tests pass.

Skills

Multi-harness agent operations | dynamic-swarm orchestration | the Omni Loop

Python | TypeScript | JavaScript | Rust | weight-space models | React | Tauri | SQLite

Additional Experience

Robert H. Lord Company (RHL) - AI Concept Discussion & Warehouse-Board Demo | Sep 2026-present

David Cooke Plaster Co. - Plaster Work | Summer 2026

Dunkin' | 2024-2026

Tommy's Pizza | Summer 2024

Education

Portland High School | Portland, Connecticut | Graduated 2026

Coursework: Cybersecurity; AP Computer Science